

Network Position and Academic Performance

What We Found So Far

ECC3841 Network Economics — Presentation 2, Progress Report

Where we left off

Presentation 1 asked one question: does a student's position in a friendship network predict their grades, on top of who their direct friends are and on top of raw popularity? The plan was to measure Katz–Bonacich centrality across a range of the decay parameter ϕ , and test it while holding friend-group GPA and popularity fixed. This brief reports what that test actually found, including a gap in our own design that we found and fixed along the way.

What we did

Four steps, each building on the last.

1. Replicate Smirnov and Thurner's two checks, to confirm the data behaves the way they reported.
2. Run centrality on its own, without controlling for popularity: the naive test.
3. Add popularity as a control and sweep ϕ , pooling all five student groups: the test planned in Presentation 1.
4. Check that test's assumptions, find one was not actually met, and re-run the confirmatory test properly in the one sample where it can be.

Result 1: the replication holds

Controlling for a student's own past GPA (coefficient 0.955, school group, $n = 2,088$, $R^2 = 0.908$), a friend's past GPA does not predict that student's future GPA: coefficient -0.0055 , $p = 0.557$. New friendships show a smaller GPA gap than friendships about to end: university, -0.021 , $p < 0.001$; high school, -0.010 , $p = 0.183$, same direction, not significant. Both match Smirnov and Thurner. This is the foundation the rest of the brief builds on.

Result 2: the naive centrality effect is popularity, relabelled

Katz–Bonacich on its own, net of contact GPA, pooled across all five groups ($n = 36,696$): coefficient $+0.089$ (standardized), $p < 0.001$. Looks like position matters. Add raw in-degree and out-degree to the same regression: the Katz coefficient is $+0.012$, $p = 0.760$, while direct degrees remain significant. The naive association does not survive the degree controls.

Result 3: pooled exploration, and a gap in it

Sweeping ϕ , net of contact GPA and popularity, pooled across all five groups:

ϕ (frac. of ceiling)	0.05	0.20	0.40	0.60	0.80	0.95
Katz coefficient	+0.103	+0.092	+0.054	+0.024	+0.007	+0.001
p -value	< 0.001	0.022	0.214	0.485	0.767	0.946

At $\phi = 0.85$: coefficient +0.004, $p = 0.835$, $n = 36,696$. This pooled cross-sectional association is descriptive only, not the headline estimate.

But this regression never controlled for a student’s own past GPA, and it should have. GPA is time-varying only for the 655 high-school students (6 measurements over time). For the four university cohorts, `data.mat` stores a single, static GPA value per student — not a column that happens to be constant, but literally one column. Own past GPA cannot be built as a separate regressor for over 80% of these 36,696 observations. Presentation 1’s model equation promised this control. The test we actually ran did not have it.

Result 4: the corrected test

Restrict the confirmatory test to the one sample where a genuine $t \rightarrow t + 1$ design with an own-GPA control is possible: high school, $n = 2,088$ (535 students, up to 5 transitions each).

Before trusting it, we checked whether adding own past GPA collides with the other regressors. It does not: it correlates 0.008 with Katz, 0.065 with in-degree, 0.122 with out-degree, 0.285 with friend GPA; variance inflation factor 1.11. (Katz and in-degree still correlate at 0.866, VIF ≈ 4 — expected, and exactly what sweeping ϕ is for.)

ϕ (frac. of ceiling)	0.05	0.20	0.40	0.60	0.80	0.95
Katz coefficient	+0.006	+0.015	+0.018	+0.015	+0.010	+0.007
p -value	0.58	0.17	0.09	0.10	0.13	0.18

At the primary $\phi = 0.85$ setting, the school estimate is +0.009 ($p = 0.139$). Its 95% confidence interval is $[-0.003, 0.022]$, and no point in the ϕ sweep is statistically significant. Once a student’s own trajectory is controlled, we do not find evidence that Katz–Bonacich centrality predicts next-period GPA.

Net of popularity, contact GPA, a student’s own past performance, and common snapshot shocks, we do not find evidence that outgoing network reach independently predicts grades.

Why this is still worth reporting

Three specifications, three different pictures: naive centrality looked positive, the pooled sweep changed with ϕ , and the corrected test finds no statistically significant association. The first two looked like findings and neither survived scrutiny. That progression is itself the result:

- The high-school panel permits a real time lag and an own-performance control, which cross-sectional comparisons cannot provide.
- It shows a general risk: pooling groups with different data structures (GPA that varies over time in one, static in four) can manufacture a significant coefficient that the one correctly specified subsample does not support.
- It is a decision-relevant answer: interventions built around widening students’ network reach are not supported by this data once direct ties, contact GPA, past performance, and common time shocks are controlled.

Limitations and what's next

- The corrected test has $n = 2,088$ from 535 students — a much wider confidence interval than the pooled test's 36,696.
- It cannot be extended to the university cohorts: their GPA is a single measurement, not a time series, so no lagged design is possible for them with this data.
- The earlier pattern — the effect concentrated in sparser networks — was found under the pooled specification, without an own-GPA control. It has not been re-checked under the corrected specification and should be treated as unconfirmed.
- The main result is robust to interpreting centrality as incoming status and to retaining only reciprocal ties; these checks are less precise because they use fewer valid observations.

Full study setup, the model, and the identification strategy: [research_brief.pdf](#). This brief reports what running that plan actually found.